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An Integrated Phenotypic and Genotypic Approach Reveals a High-Risk Subtype Association for *EBF3* Missense Variants Affecting the Zinc Finger Domain

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Objective: Collier/Olf/EBF (COE) transcription factors have distinct expression patterns in the developing and mature nervous system. To date, a neurological disease association has been conclusively established for only the *Early B-cell Factor-3* (*EBF3*) COE family member through the identification of heterozygous loss-of-function variants in individuals with autism spectrum/neurodevelopmental disorders (NDD). Here, we identify a symptom severity risk association with missense variants primarily disrupting the zinc finger domain (ZNF) in *EBF3*-related NDD.
Methods: A phenotypic assessment of 41 individuals was combined with a literature meta-analysis for a total of 83 individuals diagnosed with *EBF3*-related NDD. Quantitative diagnostic phenotypic and symptom severity scales were developed to compare *EBF3* variant type and location to identify genotype–phenotype correlations. To stratify the effects of *EBF3* variants disrupting either the DNA-binding domain (DBD) or the ZNF, we used *in vivo* fruit fly UAS-GAL4 expression and *in vitro* luciferase assays.
Results: We show that patient symptom severity correlates with *EBF3* missense variants perturbing the ZNF, which is a key protein domain required for stabilizing the interaction between EBF3 and the target DNA sequence. We found that ZNF-associated variants failed to restore viability in the fruit fly and impaired transcriptional activation. However, the recurrent variant EBF3 p.Arg209Trp in the DBD is capable of partially rescuing viability in the fly and preserved transcriptional activation.
Interpretation: We describe a symptom severity risk association with ZNF perturbations and EBF3 loss-of-function in the largest reported cohort to date of EBF3-related NDD patients. This analysis should have potential predictive clinical value for newly identified patients with *EBF3* gene variants.

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Early B-Cell Factor 3 (*EBF3*) encodes a Collier/Olf/EBF-(COE)-transcription factor that controls gene expression via DNA methylation and chromatin modifications.[1–3] COE-transcription factors are composed of an N-terminal DNA-binding domain (DBD) with a zinc finger domain (ZNF) required to stabilize DNA-binding, an immunoglobulin-like, plexins, transcription factors (IPT/TIG) domain, and a C-terminal helix–loop–helix (HLH) required for dimerization.[4] These regions are >60% identical across invertebrate and vertebrate COE-transcription factors.[4,5] EBF3 contributes to nervous system, immune system, bone, gastrointestinal tract, and muscle development.[4,6–9] Loss-of-function (LOF) manipulations of *EBF3* homologs in worms (*unc-3*, *CeO/E*), flies (*knot (kn)*, *collier*), frogs (*Xcoe2*), and mice (*Ebf3*) show that complete EBF3 LOF is deleterious to survival and impairs neurodevelopmental processes.[4] In the mouse nervous system, *Ebf3* regulates neuronal proliferation, cell-type specification, and migration,[4,8,10,11] indicating that EBF3 is a key transcriptional regulator of nervous system development and function. These findings are consistent with a prior *in vitro* study revealing that EBF3 binds to the promoters of many genes associated with neurodevelopmental disorders (NDD).[12] Despite these roles, the potential impact of EBF3 LOF in human disease was unknown until the discovery of *EBF3* variants associated with NDD and autism spectrum disorders.

Heterozygous genomic deletions disrupting the distal arm of chromosome-10, predominantly involving the *EBF3* genomic locus, cause 10q26-deletion syndrome [MIM:609625]. This syndrome is characterized by delayed development, hypotonia, ataxia, skeletal abnormalities, short stature, and neuropsychiatric comorbidities.[13–17] In 2016, heterozygous *EBF3* variants and single gene deletions were found to cause the autosomal dominant Hypotonia, Ataxia, and Delayed Development (HADD) syndrome [MIM:617330], which is characterized by its namesake features and co-morbidities, including autism.[5,12,18,19] These overlapping features indicate that EBF3 (LOF) is a causative factor for both conditions. Although the discovery of a monogenic disorder associated with *EBF3* is fairly recent, increasing evidence suggests that *EBF3*-related NDD may be more common than previously recognized,[19–24] with *EBF3* identified as the causative gene in 1:1,200 individuals with NDD.[25]

Given the recent identification of *EBF3*-related NDD, the full phenotypic spectrum and molecular association with the symptom profile and disease severity remains unknown. Improved understanding of the phenotypic and genotypic correlations would facilitate clinical recognition, prognostication, and mechanistic understanding. Here, we combine a comprehensive phenotypic and genotypic analysis of 41 individuals diagnosed with *EBF3*-related NDD with a literature meta-analysis for a combined assessment of 83 individuals. We present the clinical data for these individuals, describe quantitative phenotypic scales to identify an individual's risk for a pathogenic *EBF3* variant and quantify symptom severity, and determine genotype–phenotype correlations that reveal a symptom severity risk association with a conserved protein domain.

Materials and Methods

Study Participants

This study was approved by the institutional review boards of the participating institutions. Study participants were identified through self-referrals by families, GeneMatcher,[26] or personal communication from colleagues. The study includes individuals of both genders across the lifespan. Inclusion criteria include expert clinical evaluation and molecular confirmation of a pathogenic *EBF3* variant or gene deletion by chromosomal microarray or exome sequencing. Exclusion criteria include other co-morbid pathogenic gene alterations and brain injury due to trauma, infection, or stroke. Informed consent for study participation is obtained from adult participants and/or a parent or legal guardian. Assent is obtained from participants age 7-years and older who are cognitively able to provide agreement. All data were de-identified. The study includes 41 individuals with an average age at enrollment of 8.5 years old (age range: 16 months old to 52-years old). Three individuals opted out of publishing genomic information (probands 9, 30, and 33).

We conducted a phenotypic and genotypic analysis of 41 individuals with *EBF3*-related NDD and correlated the findings with a meta-analysis of 47 individuals.[5,12,18–24,27–31] Five individuals in our study were previously reported.[5,12,21,22,31] Probands 7, 15, 19, 25, and 26 from our study (Supporting Information Table S1, which is available online) are the same as meta-analysis probands M-11, M-1, M-25, M-31, and M-32, respectively (Supporting Information Table S2). Together, our study cohort and the meta-analysis encompass 83 individuals. Developmental and medical history was obtained via caregiver report, clinician assessment, and review of medical record. We analyzed and compared demographics, clinical phenotypes, and molecular findings in our cohort to the meta-analysis (Supporting Information Tables S1 and S2). Variant nomenclature is according to the gene transcript (GenBank:NM_001005463.3, GRCh37). For comparison to a control NDD population, we applied the same comprehensive phenotypic assessment for 9 unrelated individuals without *EBF3* alterations (Supporting Information Table S3).

For height, weight, and occipito-frontal circumference (OFC) measurements, we calculated LMS (lambda-mu-sigma)-based standard z-scores,[32] where for a given L (degree of skewness), M (median), and S (variation coefficient) values, and an experimental value X, the z-score $= [(X/M)^L - 1]/(LS)$. L, M, and S values for height and weight based on sex and age (birth to 20 years) are from the Centers for Disease Control and Prevention (CDC).[33] For OFC, the L, M, and S values are from the World Health Organization (WHO, birth to 5 years).[34] For individuals born prematurely, growth parameter z-scores were

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calculated using the mean and standard deviation from established statistics on premature infants.[35] For example, if a participant was born at 36 weeks gestational age (GA) with a birth weight of $x = 2.7$ kg, then we take the average and standard deviation for birth weight of 36-week GA infants ($\mu = 2.65$ kg and $\sigma = 0.509$ kg, respectively), and calculate $z = (x-\mu)/\sigma = 0.098$. Delayed development is determined by failure to achieve developmental milestones within the CDC's standard age range (Supporting Information Tables S4–S6).[36]

EBF3-DBS Assessment Algorithm

EBF3-Developmental and Behavioral Scale (EBF3-DBS) was developed to determine an individual's risk of a pathogenic *EBF3* variant via a quantitative human phenotype score. It uses a subset of the 71 features, each weighted based on its frequency in the study cohort to determine a score from 0 to 10, with 10 representing the highest risk for a pathogenic *EBF3* variant. Operational definitions of the features are provided in Supporting Information Table S7. We used 2-fold cross-validation to test the algorithm's ability to assign a score to phenotypic data that were not used in estimating the phenotypic feature prevalence.[35] Cross-validation is a resampling method that uses different portions of the data to test and train the algorithm on different trials and to assess how the results will generalize to an independent data set. We compared our study cohort to both the NDD cohort with molecular testing excluding *EBF3*-related diagnoses (Non-EBF3 NDD) and the meta-analysis of *EBF3*-related NDD (EBF3 Literature). The study cohort was randomized into Groups 1 and 2 for 2 trials. In Trial 1, we trained the algorithm on Group 2's feature prevalence and tested the algorithm in Group 1, Non-EBF3 NDD, and EBF3 Literature. In Trial 2, we trained the algorithm on Group 1's feature prevalence and tested the algorithm in Group 2, Non-EBF3 NDD, and EBF3 Literature.

For the 2-fold cross-validation, we assigned feature frequencies in the range of [25%, 50%) with "["indicating inclusion of 25% and ")" indicating exclusion of 50%, [50%, 75%), [75%, 100%), and [100%] weighted scores of 0.25, 0.5, 0.75, and 1, respectively. If the features appear in <25% of the study cohort randomized to Group 1 or Group 2, then they are weighted 0. The EBF3-DBS score when including all 71 features is calculated as: W(x) = weight of a feature x (0, 0.25, 0.50, 0.75, or 1); F(p) = set of clinical features identified for individual *p*; M = sum of the feature weights = 17 for Trial 1 and 17.25 for Trial 2. The EBF3-DBS score for the most commonly reported features (occurring ≥75% frequency in the study cohort) is calculated as: W(x) = weight of a feature x (0.75 or 1); F(p) = set of clinical features with ≥75% reported frequency identified for individual *p*; M = sum of the weights of features with >75% reported frequency = 7 for Trial 1 and 8.25 for Trial 2.

EBF3-DBS score for individual *p* is calculated as:

$$EBF3-DBS(p) = \left\lceil \frac{10}{M} \sum_{x \in F(p)} W(x) \right\rceil$$

EBF3-DDSS Assessment Algorithm

EBF3 Developmental Delay Severity Scale (EBF3-DDSS) is a quantification of phenotypic severity based on features occurring between 50–75% in the study cohort and the average number of months delayed across a subset of motor milestones including holding one's head up when prone, sitting independently, pulling to stand, cruising, and walking (Supporting Information Table S6).[34] The milestones were selected due to the prevalence of motor developmental delays and were most frequently reported by caregivers. If the individual was born prematurely, then we corrected for GA (full-term = 38-weeks GA).

EBF3-DDSS integrates the degree of motor developmental delay with the presence or absence of phenotypic features observed at a prevalence of 50–75% to derive a single numeric score. A higher score is associated with greater motor developmental delays and co-morbidities. EBF3-DDSS score is calculated as follows:

X(p) = feature sum calculated by adding 1 point for each feature present and subtracting 1 point for each feature absent for individual *p*. *Y(p)* = the average months delayed across the selected motor milestones for individual *p*.

EBF3-DDSS score for individual *p* is calculated as:

$$EBF3-DDSS(p) = Y(p)*1.3^{X(p)}$$

To assess the EBF3-DDSS efficacy, we used 2-fold cross-validation[35] of the combined 83 individuals to compare the scores of individuals with coding variants within the ZNF (Coding, ZNF), coding variants outside of the ZNF (Coding, Non-ZNF), and noncoding variants (Noncoding). We randomized the individuals into 2 groups, 1 and 2, and conducted 2 trials. In Trial 1, we trained the algorithm using the features occurring 50–75% in Group 2 and tested it on Group 1. In Trial 2, we trained the algorithm with Group 1 and tested it on Group 2. The scores from each test trial were separated based on the 3 variant classes. Individuals included in the testing groups were determined by permission to publish genomic information and availability of caregiver reported motor developmental milestones.

In Silico Prediction Algorithms

The predicted impact of *EBF3* variants was obtained using Protein Variation Effect Analyzer (PROVEAN) to predict whether an amino acid substitution or indel has an impact on the biological function of a protein,[37] Combined Annotation Dependent Depletion (CADD) to score the deleteriousness of single nucleotide variants and indels,[38] and Genomic Evolutionary Rate Profiling (GERP) to identify position-specific estimates of evolutionary constraint.[39] The NCBI reference sequences NP_001005463.1 and NM_001005463.3 were used.

Fruit Fly Functional Assessment

To determine the *in vivo* functional consequences of *EBF3* variants, we used the Upstream activating sequence (UAS)-GAL4 system to express wild-type (WT) and variant *EBF3* in the fruit fly *Drosophila melanogaster* as described.[5] *kn-t2A-GAL4* drives the expression of UAS-cDNA in the endogenous spatiotemporal

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pattern of the fly gene *kn*[5]. We used the published UAS-EBF3 fly alleles for EBF3 WT, p.R163Q, and p.R163L.[5] We generated UAS-cDNA fly alleles with a full-length cDNA for *EBF3* (GenBank:HQ258299) containing a C-terminal hemagglutinin (HA) tag for variants p.G171D, p.R209W, and p.H157_C170del to delete the ZNF domain (p.ΔZNF). Site-directed mutagenesis was performed as described.[5] *kn-t2A-GAL4* and UAS-EBF3 cDNA alleles were used in conjunction with the deficiency allele *Df(2R)BSC429* (BDSC#24933), containing the *kn* locus.[5] Expression analysis of third instar larval wing disc was performed as described.[5] Images were acquired on a Leica Sp8 laser-scanning confocal microscope. Images were acquired as a z-stack with a z-step of 1-mm and line average of 4 at 400-Hz with a 25X water objective at 1024x1024 pixel resolution. Maximum intensity projections were created from the stack in ImageJ. Wing discs were immunolabeled with mouse-anti-HA (Covance, MMS-101P) and DAPI (Cayman Chemicals, #14285) as described.[5]

Luciferase Reporter Assay

To identify the consequences of *EBF3* variants on transcriptional activation, luciferase assays were conducted in HEK293 cells. Transcriptional activation by EBF3 was assessed as the ratio of Nanoluciferase to firefly luciferase. Generation of EBF3 p.R163L, p.R163Q, p.G171D, p.R209W, and p.H157_C170del (ΔZNF) variants was performed as described.[5] cDNAs were subcloned into the pcDNA-DEST40 expression vector and confirmed with Sanger sequencing. The previously reported pNL3.1-EBF3-DBSx6 vector with 6 concatemerized COE DNA-binding sites was used for the EBF3-dependent activation of Nanoluciferase expression.[5] HEK293 cells were plated in a 96-well microplate at a density of 1.5×10^4 cells/well and maintained in Dulbecco's modified Eagle's medium (DMEM) + 10% fetal bovine serum (FBS) and 5% penicillin/streptomycin. HEK293 cells were co-transfected with 170-ng pcDNA-DEST40 vectors, 120-ng pNL3.1-EBF3-DBSx6, and 60-ng pGL4.53 vectors 24-hours post-seeding using Lipofectamine-2000 (ThermoFisher Scientific, #11668027). For background control, we transfected only the pNL3.1-EBF3-DBSx6, pGL4.53, and pcDNA-DEST40 without cDNA vectors (empty vector). Media was removed at 36-hours post-transfection, and cells were lysed with 80-μL of Passive Lysis Buffer (Promega, E1941) at room temperature on a vortex-mixer for 20-minutes. Luciferase assay was performed with the Nano-Glo® Dual-Luciferase® Reporter Assay System (Promega, N1610). Luminescence was read with the BMG LABTECH FLUOstar Optima.

Results

Demographics, Growth Parameters, and Inheritance in EBF3-Related NDD

We found that both genders are equally affected, 90% of the diagnoses were made at ≤18-years old, and *EBF3* pathogenic variants were discovered in multiple ethnicities (Fig. 1A–C, Supporting Information Table S1). Self-reported ethnicity included European (66%), African (5%), East Asian (7%), Hispanic (5%), Middle Eastern (5%), and multiethnic (12%) descent (Fig 1C). Growth parameters reveal significantly shorter stature and lower weight compared to OFC at age of evaluation (Fig 1D, Supporting Information Table S1).

The inheritance pattern of pathogenic *EBF3* variants was *de novo* in 66% of individuals (Fig 1E). Prior studies identified Mendelian inheritance of a pathogenic *EBF3* variant in 1 affected mother and son pair,[24] and inheritance of a pathogenic *EBF3* variant from an unaffected parent with mosaicism.[12] In our cohort, we identified Mendelian inheritance in 5 affected individuals from 4 unrelated families and mosaicism in the unaffected parents of 2 affected and unrelated individuals (Fig 1E). Comparative analysis of CADD[38] and GERP[39] scores between our study cohort and the meta-analysis reveal that the *EBF3* variants primarily have CADD scores >20 and GERP scores >4, which are associated with higher pathogenicity prediction (Fig 1F).

Prevalence and Location of EBF3 Gene Variants and Genomic Deletions

In our study cohort, we identified 7 individuals with noncoding *EBF3* variants, 2 individuals with 10q26-deletion disrupting the *EBF3* locus (Fig 2A), and 32 individuals with coding *EBF3* variants (Fig 2B). By comparison, 5 individuals with noncoding *EBF3* variants, 5 individuals with 10q26-deletion disrupting the *EBF3* locus (Fig 2A), and 37 individuals with coding *EBF3* variants (Fig 2B) were identified in the meta-analysis.

The majority of *EBF3* variants cluster within the N-terminal DBD and are enriched within 5 amino acids of the ZNF (p.His157-Cys170) (Fig 2A,B), which is required for stabilizing the interaction of EBF3 with the DNA target.[40] Intriguingly, we identified that the most frequent *EBF3* variants affected the p.Arg163 codon with the occurrence of p.Arg163Gln (c.488G > A) in 4, p.Arg163Pro (c.488G > C) in 2, p.Arg163Trp (c.487C > T) in 2, and p.Arg163Leu (c.488G > T) in 1 individual. All 9 individuals are unrelated and together represent 10% of individuals with pathogenic *EBF3* variants.

We identified other recurrent variants affecting the p.Arg209 codon in 9, p.Arg206 codon in 5, p.Cys161 codon in 3, and p.His157 codon in 4 individuals. Both p.Cys161 and p.His157 are located in the ZNF domain and together represent 8% of the identified *EBF3* variants. The p.Arg209 and p.Arg206 residues are located in the DBD distal to the ZNF domain and together represent 17% of the identified *EBF3* variants. These recurrent *de novo* variants affecting the p.Arg163, p.R209. p.R206, p. Cys161 and p.His157 codons may be due to the position of the affected nucleotides in CpG-dinucleotide islands, which are mutational hotspots underlying over one-third

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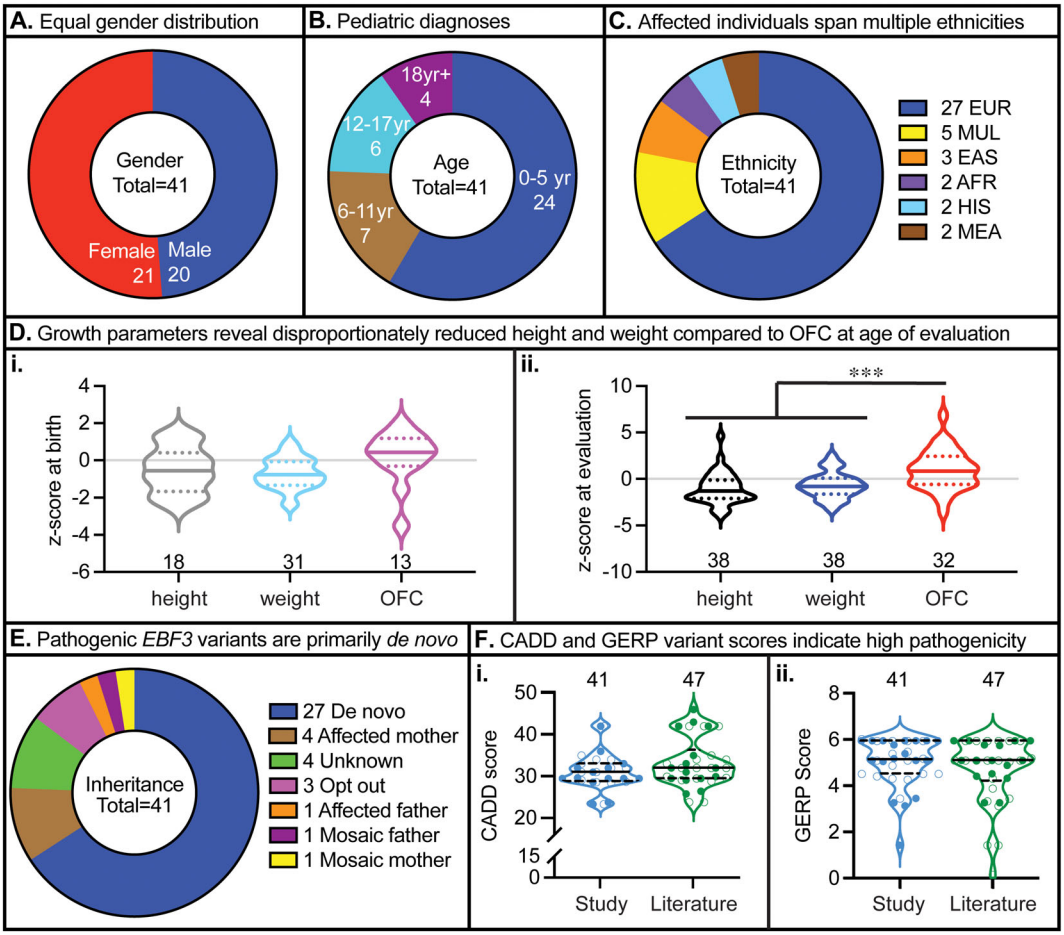

FIGURE 1: Demographics of individuals with *EBF3*-related NDD. Comparison of demographics, growth parameters, and variant features from the 41 individuals in this cohort. (A) Gender distribution was equivalent between our cohort and the meta-analysis. (B) The molecularly confirmed diagnosis of *EBF3*-related NDD occurred predominantly in the pediatric population, with more than 90% of affected individuals evaluated at less than 18 years old. (C) *EBF3*-related NDD diagnoses span multiple ethnicities. EUR = European, MUL = Multiethnic, EAS = East Asian, AFR = African, HIS = Hispanic, MEA = Middle Eastern. (D) Growth parameters reveal (i) proportionate height, weight, and occipitofrontal circumference (OFC) at birth, but (ii) disproportionately reduced height and weight compared to OFC at age of evaluation. Data are shown as mean ± standard error of mean (SEM). (E) Inheritance pattern for the majority of identified *EBF3* pathogenic variants was *de novo* (66%). Inheritance was observed from either affected parent (12%) or mosaic parent (5%). (F) CADD and GERP variant scores indicate high pathogenicity with CADD scores >20, and GERP++ scores >4. Findings were consistent between the study cohort of 41 individuals and the meta-analysis of 47 individuals. Sample size, *n*, of affected individuals is shown in the graphs. Statistical significance is calculated by 1-way analysis of variance with Tukey's multiple comparisons test, ***$p < 0.001$.

of *de novo* missense variants associated with human diseases.[41–45] Other potential mechanisms include the selfish spermatogonial selection process,[46] the error-prone replication hypothesis,[47] or a selection bias for similar phenotypes.

Neurologic Features in EBF3*-Related NDD*

Delayed language and motor developmental milestones are frequent findings in *EBF3*-related NDD (Fig 3A,B, Supporting Information Tables S4, S5). Over 97% of individuals in our study cohort have speech delay (Fig 3C). Although, 35% of individuals are able to follow 1-step and 2-step verbal commands on time, we found that over 85% of individuals have delays in expressive language (Fig 3A). This discordance in expressive and receptive language development may be partly due to motor incoordination arising from cerebellar dysfunction.[48,49] Other common speech findings in our cohort include weak or absent cry during infancy (90%), nonverbal (37%), dysarthria (27%), and speech apraxia (24%) (Fig 3C).

Prevalent motor features in our cohort include hypotonia (100%), delayed motor development (98%), ataxia (78%), gait ataxia (73%), incoordination (63%), hypomimia (63%), hypertonia (39%), non-ambulatory (34%), and hypertonia in the lower extremities (27%) (Fig 3D). Additional neurologic features include 34% of individuals with an abnormality present on electroencephalograms

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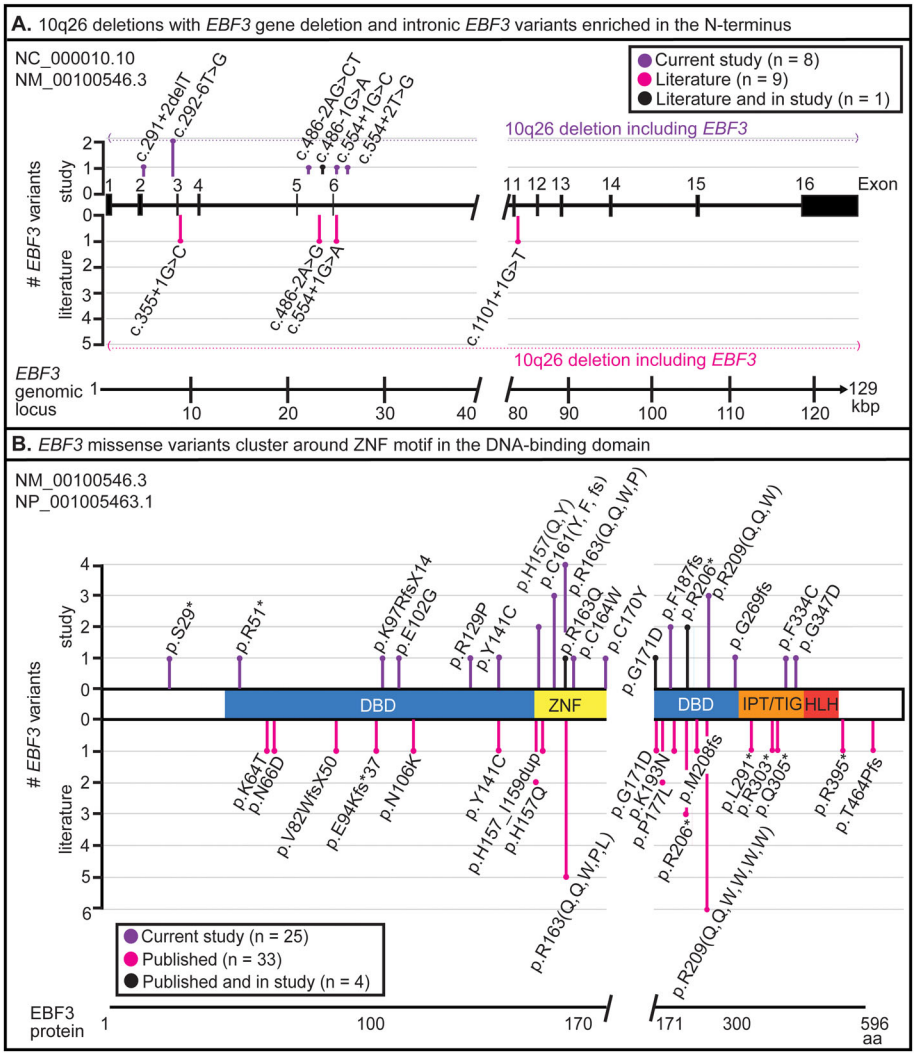

FIGURE 2: *EBF3* variants and gene deletions identified in individuals with *EBF3*-related HADD syndrome or 10q26-deletion syndrome. (A) Heterozygous 10q26-deletions containing *EBF3* and intronic *EBF3* variants were identified in 9 individuals in the study cohort and 10 individuals in the literature, which represent a total of 18 unique individuals. Variants identified in the study are shown above the genomic locus schematic and variants identified in the meta-analysis are shown below the genomic locus schematic. The intronic variants are located in the introns between exons 2 through 11, encoding the DNA-binding and ZNF domains. NCBI Reference sequences: NC_000010.10 and NM_00100546.3 (B) Heterozygous *EBF3* missense variants were identified in 29 individuals in the study cohort and 37 individuals in the literature, which represent a total of 62 unique individuals. Variants identified in the study are shown above the protein schematic and variants identified in the meta-analysis are shown below the genomic locus schematic. The *EBF3* missense variants cluster predominantly in the DBD and are enriched around the ZNF domain. Published variants are shown in magenta, variants unique to this study are in purple, and variants shared between our cohort and the literature are in black. DBD is the DNA binding domain, ZNF is the zinc finger motif, IPT/TIG stands for immunoglobin-like, plexins, transcription factors/transcription factor immunoglobin fold structure, and HLH is the helix–loop–helix region. Sample sizes are shown in each graph.

(EEG), 20% with a history of febrile seizures, but only 15% with unprovoked seizures (Fig 3E).

We found altered sensory processing to be a cardinal feature of *EBF3*-related NDD. 95% of individuals in our study had a chronic decreased pain response (Fig 3F). Other sensory features identified include repetitive itching behaviors (54%), particularly affecting the lower back (37%), increased hearing sensitivity (41%), and attraction to bright lights (20%) (Fig 3F). Neuropsychiatric comorbidities in our cohort include autistic features (68%), formally diagnosed autism (27%), attention deficit and hyperactivity disorder (ADHD, 17%), and emotional lability (17%) (Fig 3G). Common autistic features identified in our study include stereotypies (63%), poor eye contact (44%), noise aversion (53%), and aversion to crowds (enochlophobia, 17%) (Fig 3H).

Consistent with the neurodevelopmental impairments, neuroimaging studies in our cohort revealed the most frequently affected brain structure is the cerebellum

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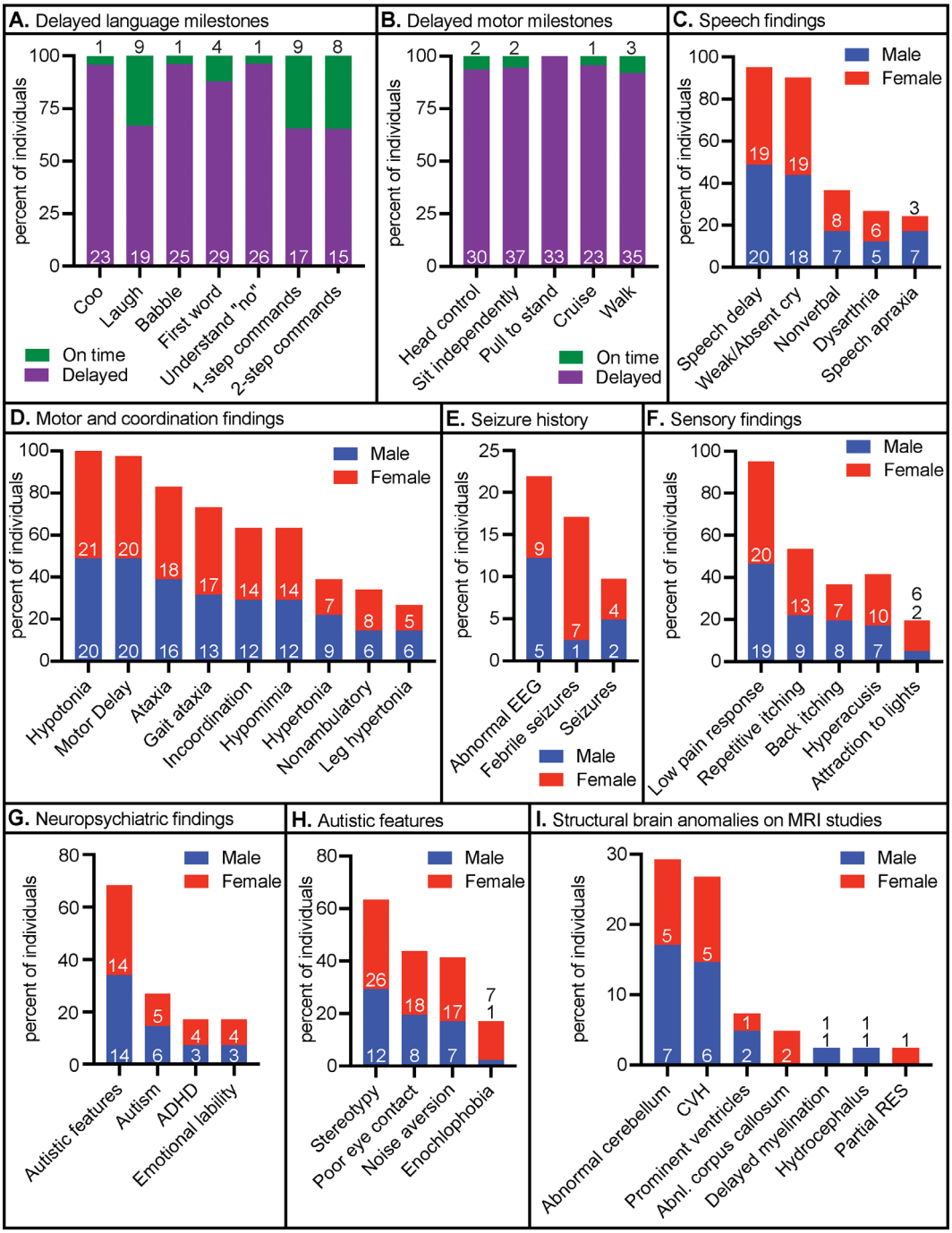

FIGURE 3: Prevalence and type of neurological features in *EBF3*-related NDD. All subjects had neurologic co-morbidities. Motor (A) and language (B) developmental delays are frequently reported. Milestones achieved on time are shown in green and milestones that are delayed are shown in purple. (C) Speech and language findings reported in >20% of affected individuals in our study cohort include speech delay, weak or absent cry, nonverbal, dysarthria, and speech apraxia. (D) Motor and coordination findings observed in >20% of affected individuals include hypotonia, motor delay, ataxia, gait ataxia, incoordination, hypomimia, hypertonia, non-ambulatory state, and lower extremity hypertonia. (E) Abnormal brain activity patterns that were observed in >5% of affected individuals include abnormal electroencephalogram (EEG), febrile seizures, and unprovoked seizures. (F) Prevalent sensory features identified in our study cohort include decreased pain response, repetitive itching behaviors, hyperacusis, and attraction to lights. (G) Prevalent neuropsychiatric comorbidities identified in our study cohort include autistic features, autism, attention deficit and hyperactivity disorder (ADHD), and emotional lability. (H) Common autistic features include stereotypies, poor eye contact, noise aversion, and aversion to crowded environments (enochlophobia). (I) Neuroanatomical findings observed in our cohort include abnormal cerebellum anatomy, cerebellar-vermian hypoplasia (CVH), prominent ventricles, abnormal (abnl) corpus callosum, delayed myelination, hydrocephalus, and partial rhomboencephalosynapsis (RES). Data are shown as percent of individuals with the feature. Sample sizes are shown above each bar. Females are shown in red and males in blue.

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(29%) with cerebellar vermian hypoplasia (CVH) as the most prevalent feature (27%) (Fig 3I). Less frequently identified structural brain findings include prominent ventricles (7%), abnormal corpus callosum, delayed myelination, hydrocephalus, and partial rhomboencephalosynapsis (RES) (Fig 3I).

Non-neurological Findings in EBF3*-Related NDD*

Non-neurologic features commonly seen in individuals with *EBF3*-related NDD include craniofacial, gastrointestinal, ophthalmologic, genitourinary, and musculoskeletal involvement. Craniofacial dysmorphisms were observed in 83% of individuals, including prominent forehead (59%), ear abnormalities (34%), mouth anomalies (34%), straight eyebrows (29%), tall forehead (17%), long face (15%), pointed chin (20%), myopathic facies (20%), triangular facies (10%), anteverted nares (7%), and short and broad chin (7%) (Fig 4A). A high frequency of constipation (88%), feeding difficulties (80%), and gastroesophageal reflux (GERD, 59%) was identified in our cohort (Fig 4B). Ophthalmologic findings were frequent, with strabismus reported in 88%, and nystagmus reported in 5% of individuals (Fig 4C). Genitourinary complications were frequently observed in both genders, with urination problems (including urine retention, urinary hesitancy, or urinary incontinence) reported by 51% of individuals (significant; Fig 4D). Urinary tract infections (UTIs) were reported in 46% and recurrent UTIs reported in 44% of individuals, with over 2-fold higher prevalence in females (Fig 4D). Structural genitourinary features in our cohort include abnormal kidney or bladder anatomy (34%), vesicoureteric reflux (15%), abnormal external genitalia (12%), and cryptorchidism (10%) (Fig 4D). Musculoskeletal dysmorphisms included clinodactyly (12%), pectus excavatum (7%), and low muscle mass (10%) (Fig 4E).

Genotype–Phenotype Correlations with the EBF3-DBS and EBF3-DDSS

To assess the accuracy of EBF3-DBS scores for individuals with limited documented features, we found that observations for ≥50% of the 71 clinical features (Table S15) was required to differentiate our EBF3 study cohort from the Non-EBF3 NDD (control) cohort (Supporting Information Fig S1A–C). However, there were no individuals in the EBF3 Literature (meta-analysis) cohort with ≥50% features reported (Supporting Information Fig S1C), which is consistent with variable phenotyping in the literature (Supporting Information Table S2). To determine if the EBF3-DBS can distinguish individuals using the most commonly reported features, we used the study cohort frequency of clinical features in the [75%, 100] and [100%] ranges to assign a diagnostic risk score from 0 to 10 based on the number of most common features identified per individual (Fig 5A,B, Table S15, and Supporting Information Tables S10–S12). A score of 10 indicates the highest risk that an individual will have a pathogenic *EBF3* alteration. Inclusion of individuals in the EBF3-DBS assessment required observations ("yes" or "no") for ≥50% of the most commonly reported features (≥75% feature prevalence). We conducted 2-fold cross-validation (Fig 5A) and compared the EBF3-DBS scores from the study cohort with the control and meta-analysis cohorts (Supporting Information Tables S11 and S12). Applying the EBF3-DBS assessment to the most commonly reported features revealed that individuals with EBF3-related NDD in the study and meta-analysis cohorts had statistically significantly higher EBF3-DBS scores than individuals from the control group (Fig 5C). However, the EBF3-DBS scores are higher in our study cohort compared to the meta-analysis, which reflects a consistent reporting of clinical features in the study cohort.

To determine whether symptom severity is associated with specific protein domains, we examined the EBF3-DDSS scores from our study cohort (Supporting Information Table S13) and the literature meta-analysis (Supporting Information Table S14) relative to the *EBF3* variant type (ZNF coding variants, Non-ZNF coding variants, and noncoding variants) (Fig 5D, 5E). A significant correlation was identified between symptom severity (higher EBF3-DDSS scores) and coding variants located within 5 amino acids of the ZNF (Fig 5E). Intriguingly, more than 40% of the individuals in the study cohort and meta-analysis have *EBF3* variants in the ZNF, which may reflect a selection bias for consistent phenotypes or increased clinical testing due to pronounced symptomology. We did not identify a significant difference between ZNF coding variants and noncoding or gene-deletion variants, which may be due to the smaller sample size in these mutation categories (Fig 5E).

To determine if there was an EBF3 protein domain association with higher EBF3-DBS or EBF3-DDSS scores, we examined the scores relative to the affected nucleotide (for coding variants) or the closest coding nucleotide (for noncoding variants). For identifying genotype–phenotype correlations with the EBF3-DBS, we used deep phenotyping and included all 71 features in the quantitative phenotypic assessment. We found that across the *EBF3* coding sequence the variants were all associated with EBF3-DBS scores ≥4, but within the ZNF domain the EBF3-DBS scores were ≥7 (Fig 6A). For the EBF3-DDSS scores, we found that variants affecting the ZNF were primarily associated with EBF3-DDSS scores ≥25 (Fig 6B). The higher EBF3-DDSS scores indicate that increased symptom severity is associated

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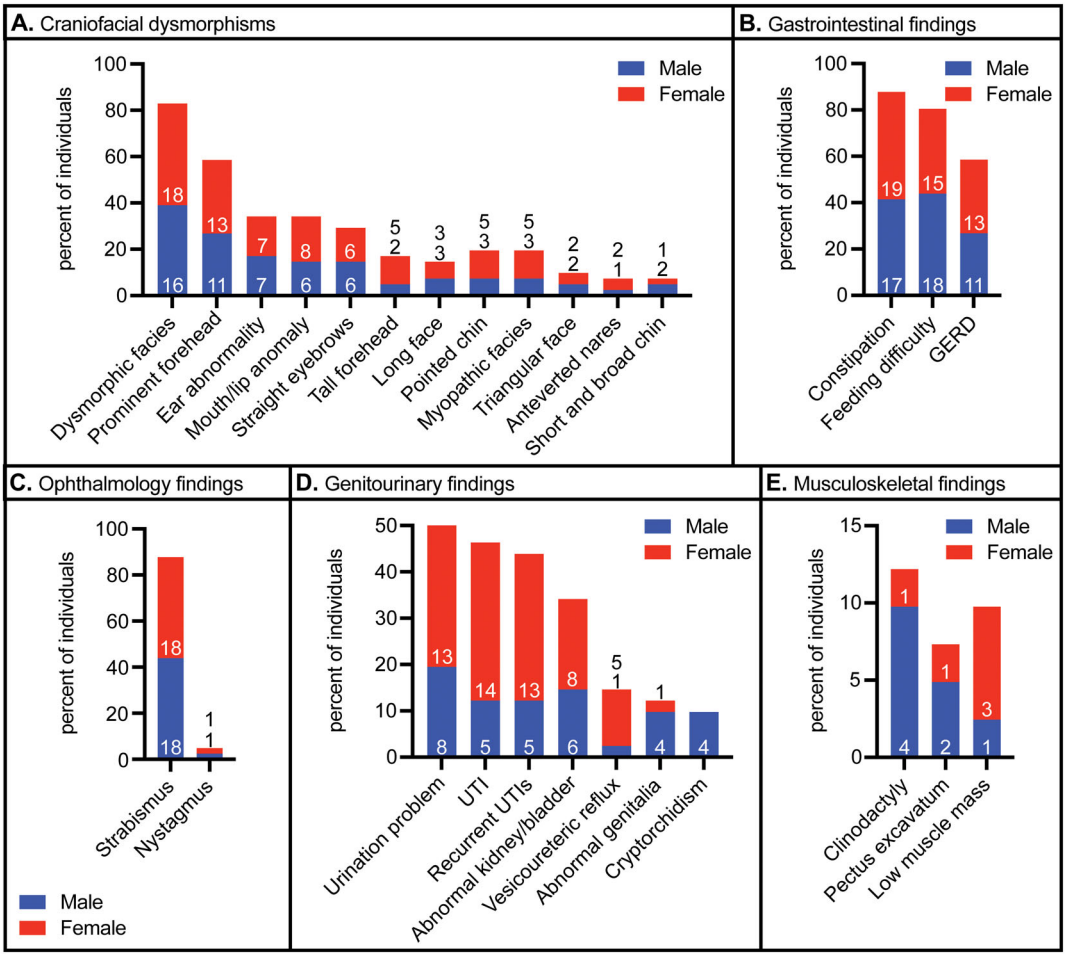

FIGURE 4: Prevalence and type of non-neurological features in *EBF3*-related NDD. Other features commonly seen in individuals with *EBF3*-related NDD include craniofacial, gastrointestinal, ophthalmologic, genitourinary, and musculoskeletal involvement. (A) Craniofacial dysmorphisms are frequently observed. In our study cohort, 84% of affected individuals were found to have at least 1 craniofacial finding including dysmorphic facies, prominent forehead, ear abnormality, mouth/lip anomaly, straight eyebrows, tall forehead, long face, pointed chin, myopathic facies, triangular face, anteverted nares, and short and broad chin. (B) Prevalent gastrointestinal findings include constipation, feeding difficulties, and gastroesophageal reflux disease (GERD). (C) Strabismus is the most common ophthalmology finding in our study cohort. Nystagmus was reported in 2 individuals. (D) Genitourinary complications are frequently observed in both genders, with urination problems and urinary tract infections (UTIs) as the most prevalent findings. Other genitourinary findings include abnormal bladder and/or kidney anatomy, vesicoureteral reflux (VUR), abnormal external genitalia, and cryptorchidism. (E) Musculoskeletal findings include clinodactyly, pectus excavatum, and low muscle mass. Data are shown as a percentage of individuals with the feature. Sample sizes are shown above each bar. Females are shown in red and males in blue.

with *EBF3* variants disrupting the ZNF. Although, the *in silico* PROVEAN analysis uniformly predicted all *EBF3* missense variants in our study cohort and the meta-analysis as deleterious (Supporting Information Tables S1 and S2), the EBF3-DDSS identified a differential symptom severity effect associated with specific protein domains. Together, our findings suggest that disrupting the ZNF is associated with more severe phenotypic consequences in *EBF3*-related NDD.

Functional Assessment of EBF3 *ZNF and DBD Variants*

To determine if there is a functional difference between *EBF3* coding variants located in the ZNF compared to the DBD, we used the strong conservation of COE-transcriptions factors in *Drosophila melanogaster* (fruit fly) to test the *in vivo* pathogenicity of the variants. In the fly, there is only 1 COE-transcription factor, *kn* (FBgn0001319), which has 62% identity and 70% similarity to *EBF3*.[5,50] Prior studies revealed that complete loss of *kn* is embryonically lethal. We previously showed that *kn-T2A-GAL4* was capable of expressing *EBF3* WT or variant cDNAs via UAS elements in the same spatiotemporal pattern as *kn* (Fig 7A).[5] Expression of EBF3 WT and Kn WT cDNAs rescued the embryonic lethality at near Mendelian expectations, but expression of the ZNF-associated EBF3 p.R163Q and p.R163L variants failed to rescue the lethality.[5] These prior *in vivo* findings revealed that the p.R163Q and p.R163L variants strongly impair EBF3 function.

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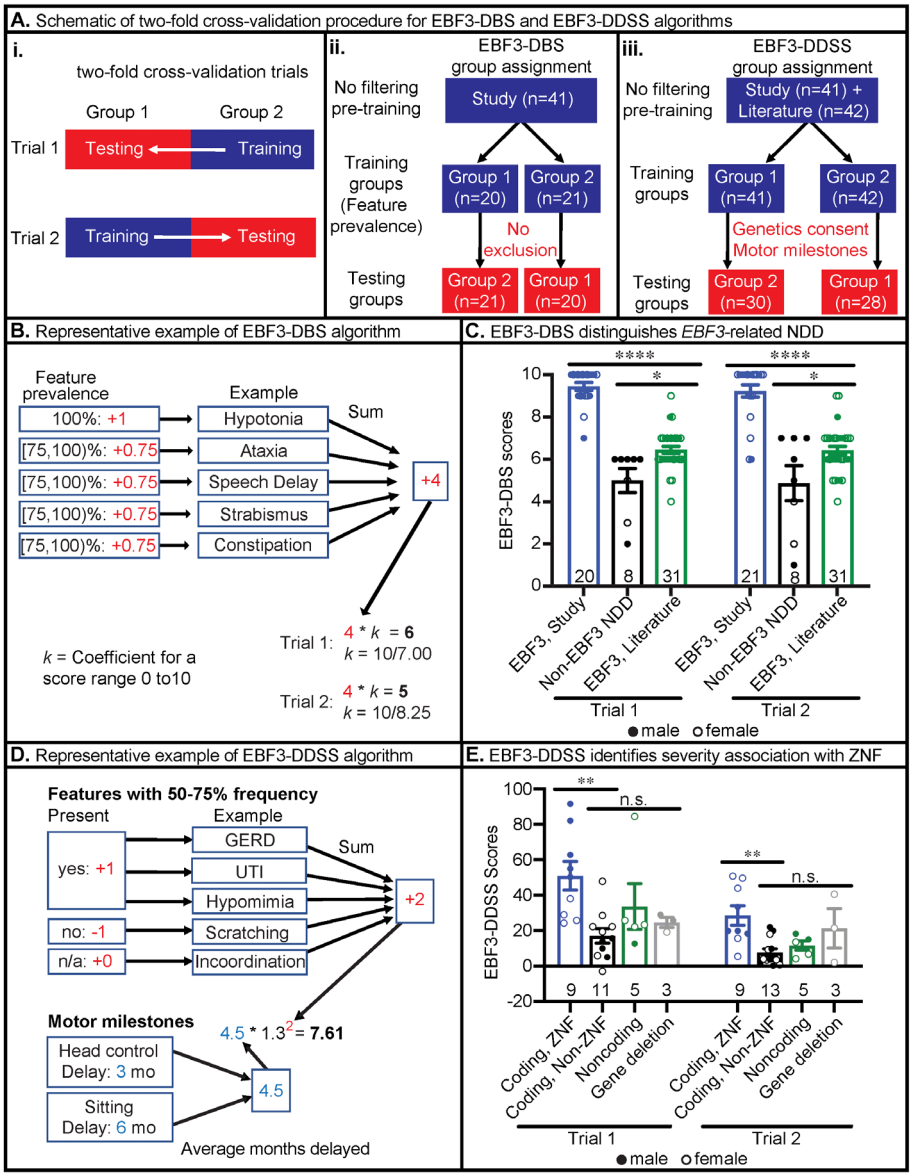

FIGURE 5: EBF3-DBS and EBF3-DDSS algorithms for identifying symptom severity risk association with *EBF3* variants. (A) Schematic of 2-fold cross-validation procedure. (i) In Trial 1, the feature prevalence in Group 2 was used to test the algorithms in Group 1. In Trial 2, the feature prevalence in Group 1 was used to test the algorithms in Group 2. (ii) For the EBF3-DBS the study cohort of 41 individuals is randomly assigned to either Group 1 or Group 2. Individuals were not excluded for the testing groups. (iii) For the EBF3-DDSS the study cohort of 41 individuals and meta-analysis (literature) of 42 individuals are randomly assigned to either Group 1 or Group 2. Individuals ($n = 3$) who opted out of publishing genomic information or did not have information about motor developmental milestones were excluded from the testing groups. (B) A representative example of the EBF3-DBS algorithm that uses the weighted prevalence of the most common phenotypic features occurring ≥75% in the study population to derive a score from 0 to 10. A higher EBF3-DBS score indicates the individual has a higher risk of having a pathogenic *EBF3* genetic alteration. (C) Individuals with ≥50% observations of the most common phenotypic features occurring ≥75% in the study population were included in the assessment. The 2-fold cross-validation of the EBF3-DBS algorithm reveal that the individuals with *EBF3*-related NDD in our study cohort (EBF3, Study) and affected individuals reported in the literature (EBF3, Literature) had a statistically higher EBF3-DBS score compared to the individuals without *EBF3* variants (Non-EBF3, NDD) in both trials. (D) A representative example of the EBF3-DDSS algorithm that uses the presence or absence of moderately common features occurring 50–75% and the average months of delayed gross motor milestones in affected individuals to derive a numerical score. Higher scores indicate increased symptom severity. (E) The 2-fold cross-validation of the EBF3-DDSS algorithm is examined relative to the *EBF3* variant type and location. A significant correlation was identified between symptom severity and coding variants within the ZNF domain (amino acids 157–170) and the flanking 5 amino acids (range 152–175) (Coding, ZNF) compared to other coding variants (Coding, Non-ZNF). Data are shown as mean ± SEM. Males are shown with solid circles and females are shown with open circles. Statistical significance is calculated by 1-way analysis of variance with Tukey's multiple comparisons test, $^{**}p < 0.001$ and $^{***}p < 0.001$. n.s. = non-significant. Sample size, *n*, of affected individuals is shown in the graphs.

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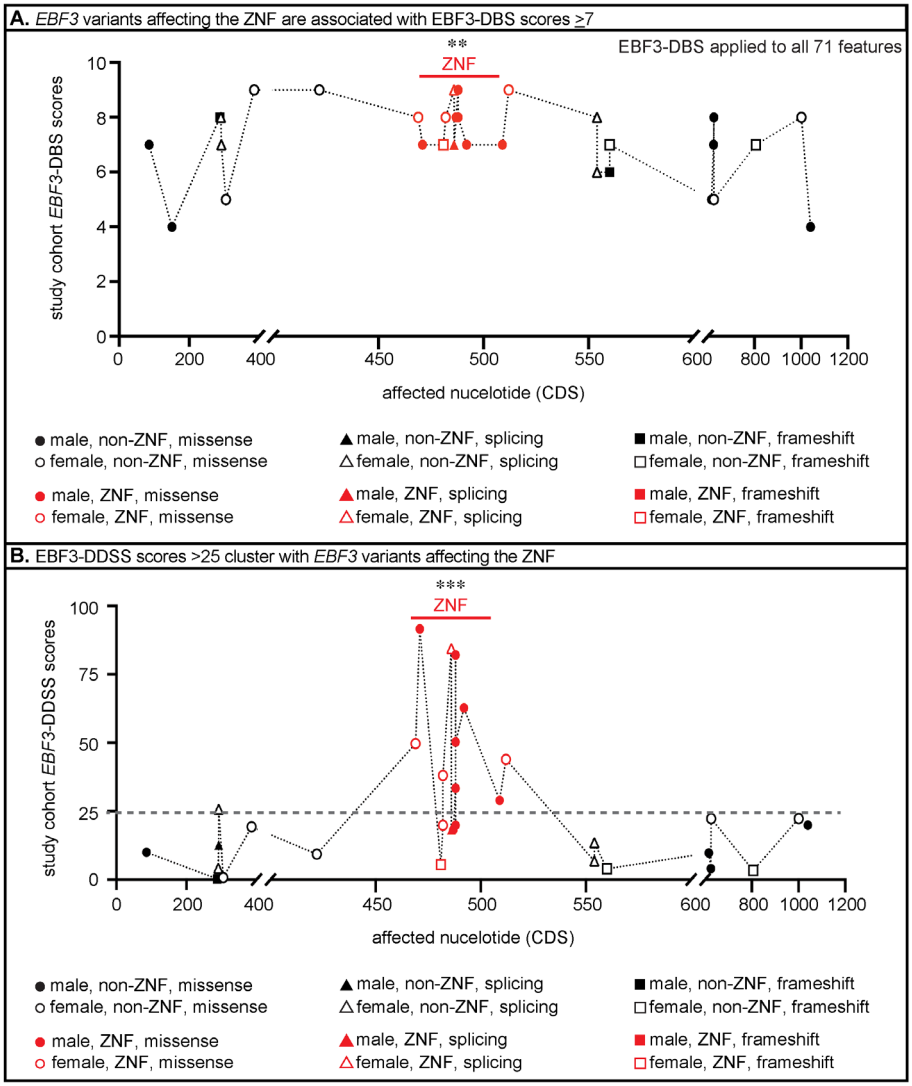

FIGURE 6: Distribution of EBF3-DBS and EBF3-DDSS scores relative to *EBF3* variant location in coding sequence. (A) EBF3-DBS scores from the study cohort using all 71 features are mapped relative to the affected coding nucleotide. Distribution of the scores reveal that variants affecting the ZNF domain (red) show an enrichment of EBF3-DBS scores ≥7. In contrast, the EBF3-DBS scores range from 4 to 9 outside of the ZNF domain (black). (B) EBF3-DDSS scores from the study cohort are mapped relative to the affected coding nucleotide. Distribution of the scores reveal that variants affecting the ZNF domain (red) show an enrichment of EBF3-DDSS scores ≥25 (dotted line). In contrast, the EBF3-DDSS scores are <25 outside of the ZNF domain (black). Comparison of EBF3-DBS and EBF3-DDSS scores affecting the ZNF domain or outside of the ZNF domain by unpaired t-test reveals statistical significance with **$p < 0.001$ and ***$p < 0.001$. Filled-in symbols are males and open symbols are females. Missense variants (circle), splicing intronic variants (triangle), frameshift variants (square).

To determine the functional difference between *EBF3* coding variants located in the ZNF compared to the DBD, we generated fly alleles for the ZNF-adjacent p.G171D variant, deletion of the entire ZNF domain (p.ΔZNF), and the recurrent DBD-associated p.R209W variant (Fig 7B). The functional impact of these 3 variants had not been previously tested. We determined that *kn-T2A-GAL4* recapitulates the endogenous expression pattern of *kn* in the third instar larval wing imaginal disc for the UAS lines encoding the HA-tagged EBF3 WT, p.G171D, p.ΔZNF, and p.R209W cDNAs (Fig 7C). The functional assessment of the *EBF3* variants by the UAS-GAL4 system was performed in conjunction with the *kn* loss of function alleles *Df(2R) BSC429* (which includes deletion of the *kn* locus) and *kn-T2A-GAL4* (Fig 7D). We scored >300 adult flies per variant and found that both EBF3 WT and Kn WT similarly rescued the embryonic lethality, which recapitulated our prior findings.[5] However, the ZNF-associated EBF3 p.G171D and EBF3 p.ΔZNF failed to rescue the embryonic lethality, similar to the ZNF-associated EBF3 p.R163Q and p.R163L variants.[5] The previously

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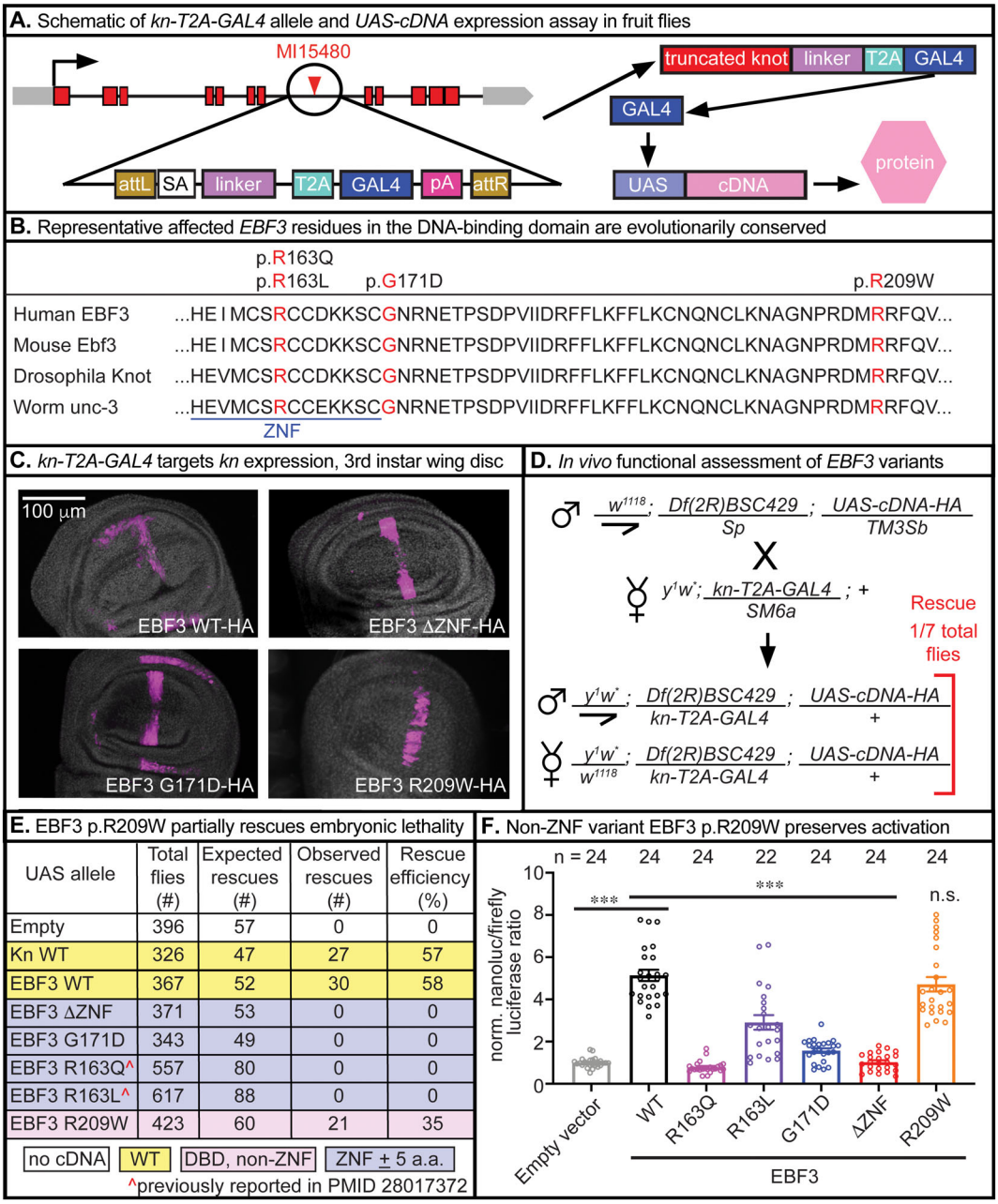

UAS allele	Total flies (#)	Expected rescues (#)	Observed rescues (#)	Rescue efficiency (%)
Empty	396	57	0	0
Kn WT	326	47	27	57
EBF3 WT	367	52	30	58
EBF3 ΔZNF	371	53	0	0
EBF3 G171D	343	49	0	0
EBF3 R163Q^	557	80	0	0
EBF3 R163L^	617	88	0	0
EBF3 R209W	423	60	21	35

FIGURE 7: Functional validation of *EBF3* variants in the ZNF and DBD. ZNF associated variants p.R163Q, p.R163L, and p.G171D and the DBD associated variant p.R209W were selected to study the differential impact of *EBF3* variants. (A) All variants are evolutionarily conserved. (B) Schematic of the UAS-GAL4 expression system using *kn-T2A-GAL4* to express *EBF3* or *knot* cDNAs in the same spatiotemporal pattern as endogenous *kn*. (C) For expression analysis, *yw/y; kn-T2A- GAL4/CyO, Kr-GAL4, UAS-GFP* males were crossed with *yw; UAS-cDNA/TM3 Sb, Kr- GAL4, UAS-GFP* virgin females, and double heterozygotes were selected by loss of GFP expression. Immunolabeling revealed that *kn-T2A-GAL4* recapitulates the endogenous *kn* expression pattern in third instar wing disc, as shown with the HA-tagged EBF3 WT, p.G171D, p.DZNF, and p.R209W. Images show HA (magenta) and DAPI (gray). Scale bar = 100 mm. (D) Fly cross schematic shows the expected rescue genotype and frequency. (E) For each *UAS-cDNA* line, >300 adult flies were scored. Data summarize the total number of flies scored, the number of expected rescue flies, the number of observed rescue flies, and the rescue efficiency. Expression of kn WT or EBF3 WT rescued embryonic lethality to viable adults. Expression of p.ΔZNF, p.G171D, p.R163Q, and p.R163L failed to rescue the lethality, such that no rescue animals were observed as adults or pupae. Rescue data for the p.R163Q and p.R163L variants were previously reported.[5] The DBD associated variant p.R209W showed partial rescue of the lethality with 35% rescue efficiency in adult flies compared to 58% rescue efficiency with EBF3 WT. (F) The ratio of Nanoluciferase/firefly luciferase was normalized to empty vector transfection control for each independent transfection. A 5-fold induction was observed with EBF3 WT and p.R209W, but the ZNF-associated variants p.R163Q, p.R163L, p.G171D, and p.ΔZNF showed poor induction. Data shown as mean ± SEM with n = 22–24 (5 to 6 technical replicates per 4 separate transfections). Statistical significance is calculated by 1-way analysis of variance with Tukey's multiple comparisons test, ***p < 0.001. n.s. = non-significant.

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reported rescue efficiency for EBF3 p.R163Q and p.R163L variants are shown for comparison (Fig 7E). In contrast, the DBD-associated EBF3 p.R209W variant rescued the embryonic lethality to viable adults at a lower efficiency compared to EBF3 WT (Fig 7E).

To assess the consequences of ZNF-associated p.G171D, p.ΔZNF, and DBD-associated p.R209W on EBF3-dependent activation of gene expression, we performed an *in vitro* luciferase assay. Transcriptional activation by these *EBF3* variants was compared to controls comprising of EBF3 WT, the previously studied p.R163Q and p.R163L variants, and empty vector.[5] The luciferase assay for EBF3 p.R163Q and p.R163L was repeated in this study for comparative analysis with the additional ZNF-associated p.G171D and ΔZNF variants, as well as the DBD-associated p.R209W variant. A 5-fold induction was observed with EBF3 WT, but the ZNF-associated variants p.R163Q, p.R163L, p.G171D, and p.ΔZNF showed poor induction (Fig 7F). In contrast, p.R209W showed a 5-fold induction similar to EBF3 WT, suggesting that this variant did not impair transcriptional activation by EBF3 (Fig 7F). Together, these findings reveal a strong deleterious effect of ZNF variants on EBF3 function.

Discussion

A common challenge faced with newly recognized or relatively rare neurodevelopmental disorders is that incomplete knowledge of distinguishing features limits the potential for prognosis and phenotypic diagnosis. We describe here the comprehensive evaluation of neurologic and non-neurologic co-morbidities in the largest reported cohort to date of affected individuals with autosomal dominant *EBF3*-related NDD, which includes 10q26-deletion and HADD syndromes. The comprehensive phenotypic profiling in our study cohort of 41 individuals with *EBF3*-related NDD revealed a spectrum of phenotypic severity with varying degrees of developmental delays and motor limitations. Despite increased information available regarding *EBF3*-related NDD, the need remains to identify the symptom severity risk factors and determine how the *EBF3* genotype affects disease presentation. To address this need, we created 2 quantitative assessment scales to determine an individual's risk for having a pathogenic *EBF3* variant (EBF3-DBS) and the degree of functional impairments (EBF3-DDSS). Integrating the EBF3-DBS and EBF3-DDSS with the affected protein domains has the potential to facilitate the accuracy of genetic testing and provide an avenue for personalized medical care.

Despite the early expression of *EBF3* in multiple developing organ systems, including the brain, there is a broad spectrum of phenotypic heterogeneity and disease severity found in association with *EBF3* variants and gene deletions. To aid clinical recognition of *EBF3*-related NDD, we developed the quantitative EBF3-DBS assessment scale to identify individuals at risk of having pathogenic *EBF3* variants and to prioritize *EBF3*-specific gene testing. The EBF3-DBS is designed to address the challenges of assessing children in various age groups and is developed for use by caregivers and healthcare providers. Existing diagnostic scales for other conditions often use criteria that are qualitatively subdivided as "major" or "minor."[51] However, a quantitative diagnostic algorithm based on phenotypic feature prevalence would reduce the subjective interpretation of "major" or "minor" criteria and could potentially be customized for application to different genetic disorders. In addition to identifying an affected individual's risk of having a pathogenic *EBF3* variant using the most commonly reported features (≥75% prevalence), the EBF3-DBS was also able to identify genotype–phenotype correlations using deep phenotyping of 71 features. When we examined the EBF3-DBS scores in conjunction with the affected *EBF3* coding sequence, we found that the EBF3-DBS scores were ≥7 for all affected individuals with ZNF-associated variants. This indicates that individuals with *EBF3* variants disrupting the ZNF have a higher prevalence of common phenotypic features compared to variants outside of the ZNF.

To determine correlations between symptom severity and *EBF3* variants, we developed the quantitative EBF3-DDSS assessment algorithm. Existing severity scales for other conditions integrate quantitative parameters with the presence or absence of various clinical features to derive a numeric score.[52] Correlating the findings from the EBF3-DDSS with *EBF3* variant type and location revealed that the degree of phenotypic severity and EBF3 LOF is linked to the specific protein domain affected by the pathogenic variant. Notably, we found that variants disrupting the ZNF were associated with greater developmental delays and co-morbidities compared to variants outside of the ZNF, which primarily affect the DBD. Interestingly, we identified that 40% of pathogenic *EBF3* variants are enriched in the codons encoding the ZNF, including the most frequently affected p.R163 codon. These findings are consistent with prior studies in the paralog *EBF1* showing that the ZNF stabilizes the interaction of COE transcription factors with the DNA target sequence and that mutating p.R163 to p.A163 abolishes DNA-binding.[40,53] Together, these findings reveal that disrupting the ZNF significantly impairs EBF3 function and lead to increased symptom severity.

To validate the predicted high-risk *EBF3* genotype–phenotype association with the ZNF domain, we

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conducted *in vivo* fruit fly lethality rescue and *in vitro* luciferase reporter transcriptional activation assays. Perturbations of the ZNF are associated with both reduced EBF3-dependent transcriptional activation and failure to restore viability in flies lacking endogenous *kn*. To examine if a differential effect exists for *EBF3* variants located outside of the ZNF, we assessed the second most commonly identified variant p.R209W, which is located in the DBD. We found that the p.R209W variant activates gene transcription comparable to EBF3 WT, and partially restores viability in the fly with a 35% rescue efficiency compared to 58% with EBF3 WT. These findings are consistent with the symptom severity risk association we identified from the EBF3-DDSS for *EBF3* variants disrupting the ZNF. The differential effect of *EBF3* variants affecting the ZNF and DBD reveals an association between the degree of EBF3 impairment and disease severity. To the best of our knowledge, this is among the first identifications of a high-risk disease association between a specific mutated EBF3 protein domain and disease severity in *EBF3*-related NDD.

In contrast to the coding variants, determining phenotype–genotype correlations for noncoding intronic variants within the *EBF3* gene body in our study is limited by the smaller sample size. The intronic *EBF3* variants identified in our cohort and meta-analysis are located within 6 nucleotides of exon-intron boundaries, with the majority occurring within 2 nucleotides. Variants located within the intronic or exonic segments of a splice site can perturb mRNA processing and result in alternative splicing that cause either nonsense-mediated decay of the mutant transcript or generation of a dysfunctional protein.[54–56] Noncoding variants in promoters and enhancers in the intergenic space can also affect gene expression via transcription factor binding, epigenetic chromatin modifications, and long-range interactions in the 3D chromatin architecture.[57,58] Intergenic variants in an *EBF3* enhancer were identified to perturb *EBF3* gene expression and contributed to milder phenotypic presentations,[59] which is consistent with our findings of a direct relationship between the degree of EBF3 dysfunction and phenotypic severity.

In conclusion, the described phenotypes in our study and their prevalence in *EBF3*-related NDD improves the characterization of the phenotypic spectrum and disease severity associated with pathogenic *EBF3* variants and whole gene deletions. The finding of a disease severity association with *EBF3* variants disrupting the ZNF is an important finding in light of emerging therapies targeting specific gene mutations in NDD. Longitudinal follow-up and new prospective recruitment efforts will further refine the *EBF3* genotype–phenotype correlations, which may lead to promising avenues for future therapeutic interventions personalized to an individual's variant risk association.

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Author Contributions

C.A.D. and H.T.C. contributed to the conception and design of the study. C.A.D., H.T.C., V.C.L., C.L.M. and J.M.P. contributed to the acquisition and analysis of data, drafting the text, and preparing figures. A.N., N.D.B., A.W.L., V.N., C.B., G.Z., E.M.V., E.B., P.J.B., M.F.W. contributed to the acquisition and analysis of the data and drafting the text.

Potential Conflicts of Interest

The authors declare no competing interests.

Data Availability

The de-identified data supporting the current study have not been deposited in a public repository, but are available from the corresponding author on request. The codes used to calculate growth parameter z-scores and to conduct the

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2-fold cross-validations are available in a public GitHub repository at https://github.com/coledeisseroth/HADDS.

Web Resources

OMIM, https://www.omim.org

GenBank, https://www.ncbi.nlm.nih.gov/genbank

Centers for Disease Control and Prevention growth charts, https://www.cdc.gov/growthcharts/index.htm

World Health Organization growth charts, https://www.who.int/childgrowth/standards/en/

gnomAD-Genome Aggregation Database, https://gnomad.broadinstitute.org/

CADD-Combined Annotation Dependent Depletion, https://cadd.gs.washington.edu/snv

UCSC Genome Browser (GRCh37/hg19), https://genome.ucsc.edu/cgi-bin/hgGateway

Protein Variation Effect Analyzer (PROVEAN), http://provean.jcvi.org/seq_submit.php

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